Multitone Frequency Generator

A Low-Cost Audio Tone Generator

Embedded Systems Lab (EC2011)

April 2025

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM

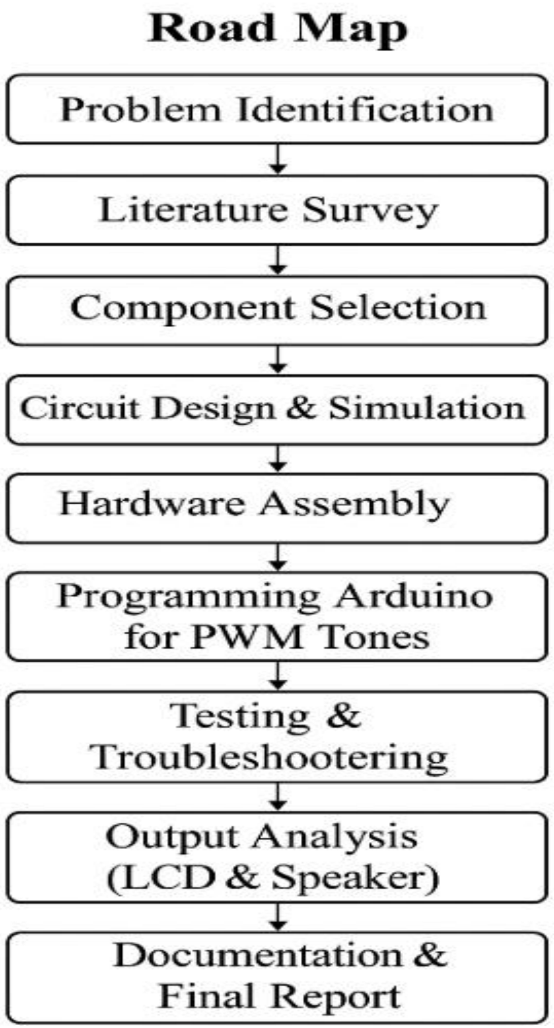
G.Deavi Asritha (EC23B1002)

N.Supriya (EC23B1011)

Ch.Pallavi (EC23B1014)

Background / Origin / Road Map

Sound is integral to electronics, serving roles in testing, education, and communication. Traditional tone generators, while effective, are often bulky and expensive, limiting accessibility for students and hobbyists. Software alternatives require additional devices, reducing portability. This project aims to develop a compact, affordable, and standalone multitone frequency generator using an Arduino platform. The roadmap includes identifying user needs, designing the circuit with audio components, building and testing the prototype, integrating real-time feedback via an OLED display, and finalizing the design for educational deployment.



Introduction

A **tone generator** produces audio signals at specific frequencies.
A **multitone generator**, as the name suggests, is capable of generating multiple tones simultaneously or sequentially. This functionality is commonly used in applications such as:
Musical Chords Generation
(Playing multiple notes together as in a basic synthesizer)
DTMF (Dual-Tone Multi-Frequency) Dialing
(Used in telephone systems to encode numbers using two simultaneous tones)
Signal Processing & Modulation Experiments
(Understanding superposition, mixing, and beat frequency concepts)

Literature Review (IEEE, ACS, Nature, Wiley)

IEEE

"PWM-Based Tone Generation Using AVR Microcontrollers" – IEEE Embedded Systems, 2020.
→ Discusses the accuracy and control of PWM-based tone synthesis.

Wiley Journal

"Digital Audio Synthesis Techniques" – Embedded Signal Processing, Wiley, 2021.
→ Describes signal mixing and frequency shaping.

ACS Circuits & Systems

"Simple Sound Generators for Educational Labs", 2022.
→ Outlines low-cost methods to generate multiple tones.

Current Scenario

Dependency on Software-Based Tools

Many tone generators are available as PC or mobile software.
These require a computer or smartphone, which limits portability and standalone operation.
Also dependent on OS compatibility and external sound cards for quality output.

Lack of Hardware-Based, Student-Friendly Devices

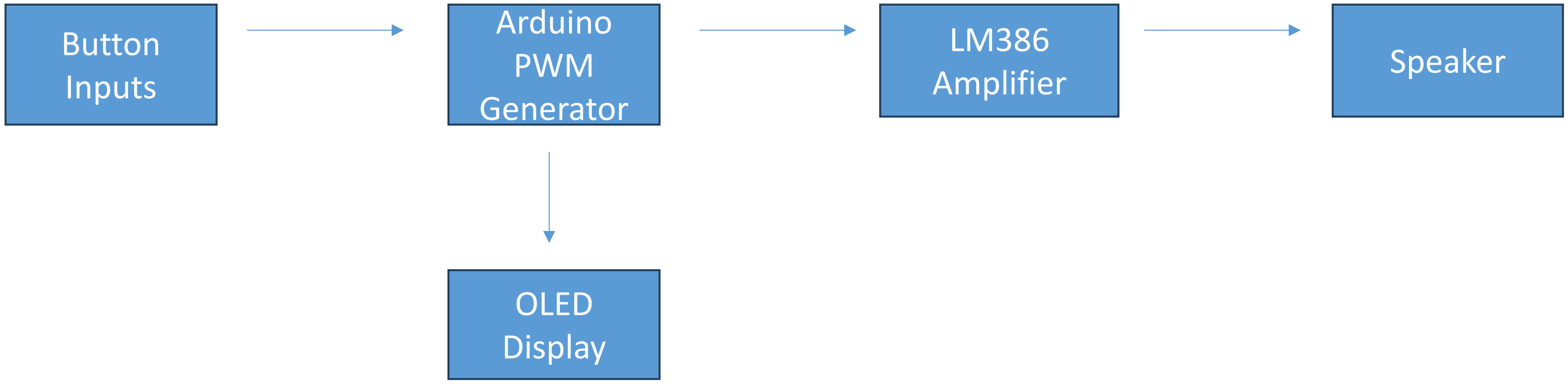
Most devices in the market are designed for professionals and not tailored for education.
No simple kits or modules for students to **learn, build, and experiment** with multitone generation.

Limited Control & Customization

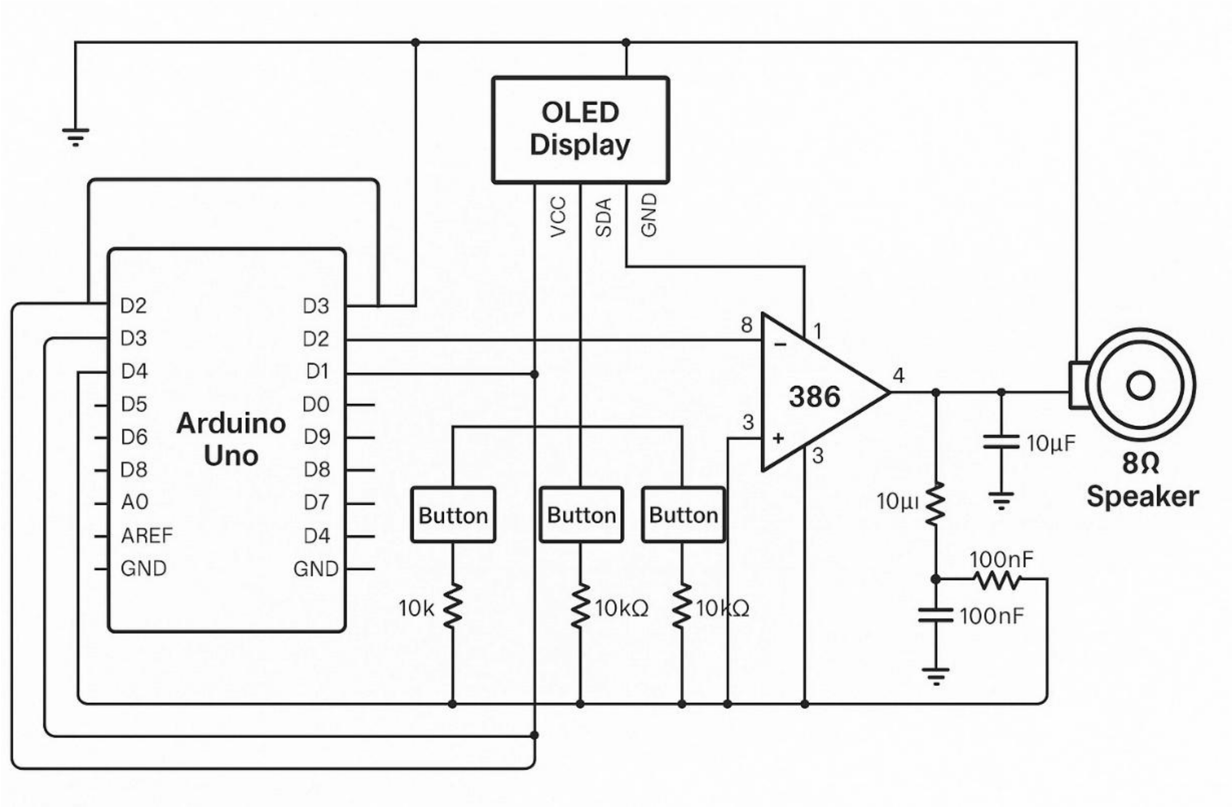
Software tools offer limited control without programming access.
Existing hardware lacks real-time frequency switching and OLED display for feedback.

Methodology / Working Principle

- **PWM (Pulse Width Modulation)** is used to simulate analog waveforms.
- Arduino rapidly switches the signal on/off to create square waves.
- Multiple buttons switch between different frequencies/tones.
- **LM386** amplifies the signal.
- **OLED** shows the current tone/frequency.
- **Power** USB or battery.

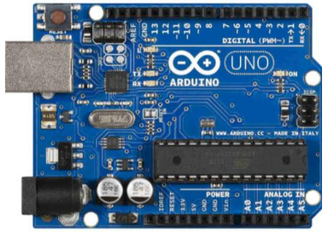


Circuit Diagram



- **Arduino Uno** generates different audio tones using **PWM**.
- **Three push buttons** select different frequencies.
- **OLED display** shows the selected tone.
- The PWM signal is sent to an **LM386 amplifier**.
- Amplified signal drives an **8Ω speaker** to produce sound.
- **Capacitors and resistors** stabilize and control gain.

<u>Components Used</u>



Arduino Uno

Push Buttons (x3)

8-ohm Speaker

Male-to-Male Jumper Wires

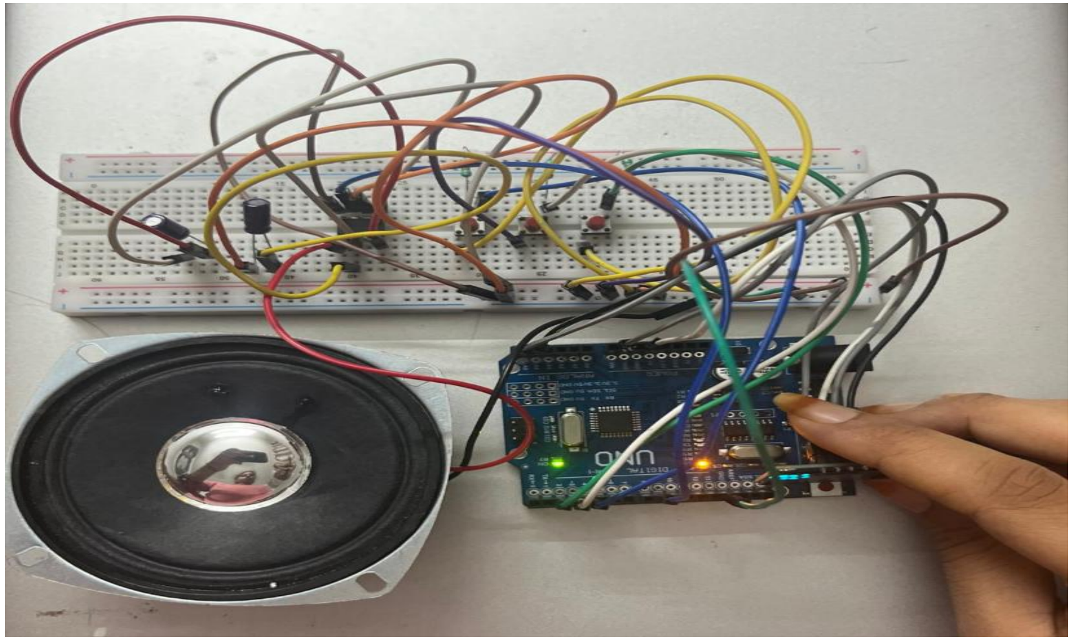
LM386 Audio Amplifier

OLED

10kΩ Resistors

10µF & 100nF Capacitors

Prototype



Results

The multitone frequency generator effectively produced *"sa re ga ma pa da ni sa" & "sa ni da pa ma ga re sa"* tones in response to button inputs, with the Arduino Uno's PWM signals accurately amplified by the LM386 to deliver clear sound through the 8-ohm speaker.
The OLED reliably displayed the selected frequencies, enhancing user interaction. Powered by a standard 5V USB source, the device operated consistently.

Conclusion

The multitone frequency generator project successfully demonstrated that a compact , and standalone device can be developed using readily available components like the Arduino Uno and LM386 amplifier. The system effectively generates multiple audio tones, provides real-time feedback via an OLED display, and operates reliably on a standard 5V USB power source. This makes it an ideal tool for educational purposes, sound experiments, and audio testing, offering an accessible alternative to traditional, expensive, and bulky equipment.

Future Enhancements

Add **rotary encoder** for better frequency selection
Replace PWM with **DAC (Digital to Analog Converter)** for smoother sound
Add **Bluetooth module** to control tone from mobile
Add **tone presets** stored in EEPROM or SD card
Expand to produce musical scales or melodies